16.2. $\delta = (r_2 - e + n_1 e) - (r_1 - e + n_2 e)$

$= r_2 - r_1 + e(n_1 - n_2)$

$= d\frac{x}{D} + (n_1 - n_2)e = 7\lambda$

$x=0$, 则 $e = \frac{7 \times 500\,nm}{0.2} = 17500\,nm = 17.5\,\mu m$.

16.3. (1) $\delta = \frac{-\lambda}{2} + (r_2 - e + ne) - r_1 = (r_2 - r_1) + (n-1)e - \frac{\lambda}{2}$

$= \frac{x}{100cm} \times 0.7mm + 0.5l - 250nm = \frac{\lambda}{2} + 3\lambda$

$x=0$ 时, $0.5l = 250nm + 250nm + 3 \times 500nm$

$l = 4000\,nm = 4\,\mu m$.

(2). $\frac{x}{100cm} \times 0.7mm + 0.5 \times 4000nm - 250nm = 0$

得 $x = -2500000\,nm$

$= -2.5\,mm$

(3). $\frac{x}{100cm} \times 0.7mm + 0.5 \times 4000nm - 250nm = 2 \times 500nm$.

$x = -1.07\,mm$.

16.5. $-\frac{\lambda}{2}+r_2-r_1=k\lambda$

$$r_2-r_1=\frac{h}{\sin\theta}-\frac{h}{\sin\theta}\cdot\cos 2\theta$$

$$=\frac{h}{\sin\theta}-\frac{h}{\sin\theta}(1-2\sin^2\theta)$$

$$=2h\sin\theta.$$

$$2h\sin\theta=(\frac{1}{2}+k)\lambda.$$

$k=0$ 时, $(\sin\theta)_{min}=\frac{\frac{1}{2}\lambda}{2h}=\frac{\lambda}{4h}$

$$\theta_{min}=\arcsin\frac{\lambda}{4h}$$

16.6. $2e\cdot\sqrt[n]{n_2^2-n_1^2}=k\lambda_1$, $k=\frac{2e}{\lambda_1}\sqrt[n]{n_2^2-n_1^2}$

$2e\sqrt[n]{n_2^2-n_1^2}=(k+\frac{1}{2})\lambda_2$, $k=\frac{2e}{\lambda_2}\sqrt[n]{n_2^2-n_1^2}-\frac{1}{2}$

$$\frac{2e}{\lambda_1}\sqrt[n]{n_2^2-n_1^2}=\frac{2e}{\lambda_2}\cdot\sqrt[n]{n_2^2-n_1^2}-\frac{1}{2}$$

$$e\cdot\left[\frac{2}{\lambda_1}-\frac{2}{\lambda_2}\right]\cdot\sqrt[n]{n_2^2-n_1^2}=-\frac{1}{2}.$$

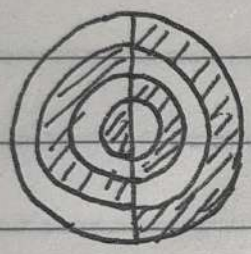
$$e=789.5nm.$$

16.9.(1). 明暗相间的平行直线.

(2). $\delta = 2ne = (7+\frac{1}{2})\lambda$.

则 $e = \frac{(7+\frac{1}{2}) \times 600nm}{2 \times 1.33} = 1.69\,\mu m$.

16.10. (1).



(2). $\delta = 2n_2 e + \frac{\lambda}{2} = (2k+1)\frac{\lambda}{2}$, $k=0,1,2\ldots$

$e_{10} = \frac{k\lambda}{2n_2} = 1.78 \times 10^{-6} m$.

$r = \sqrt{2e_{10}R} = 1.89\,mm$.

$2n_2 e = (2k+1)\frac{\lambda}{2}$. 则 $e_{10}' = \frac{(2k+1)}{4n_2}\lambda$.

$= 1.7 \times 10^{-6} m$.

暗环半径 $r' = \sqrt{2e_{10}'R} = 1.84\,mm$.